The Network Origins of Inflation Stances in a Currency Union

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Research Questions

- **Questions**
  - What structural forces shape cross-country divergence in inflation preferences in a currency union?
  - What's the welfare implication of this divergence?
- **This paper**: proposes a production-network explanation.
  - Studies optimal policy in a heterogeneous–agent input–output (HAIO) economy.
  - Defines *unilateral inflation stance* and *policy-alignment loss* (PAL).
  - Quantitatively: a country's inflation stance operates mainly through its *input–output multiplier*, and hence, relates to its position in the union's network.

Sources of Inflation Bias in the Literature

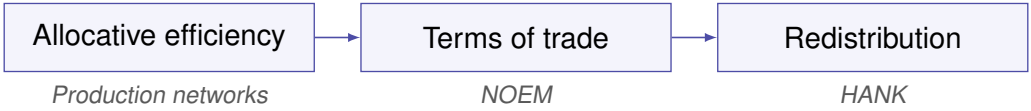
- **Terms-of-trade manipulation** in the NOEM literature ($\mathrm{ToT} \not\perp m$)
  Corsetti and Pesenti, 2001, 2005; Tille, 2001; De Paoli, 2009...
- **Redistributive motives** in the HANK literature (toward high-MU households)
  Nuño and Thomas, 2022; Dávila and Schaab, 2023.
- **Allocative efficiency** in production networks ($\mathrm{TFP} \not\perp m$)
  Baqaee and Farhi, 2020; Baqaee et al., 2024; Xu and Yu, 2025.

Which mechanism drives inflation stance in this paper?

NOEM: New Open Economy Macroeconomics. HANK: Heterogeneous-Agent New Keynesian.

Roadmap of the Theoretical Analysis

- **All three!** This paper unifies three strands of the literature.



Step 1 Derive country-level allocative efficiency in a HAIO economy.

Step 2 Show that, to 1st order, allocative efficiency arises exclusively from changes in the overall terms of trade at the efficient initial equilibrium.

Step 3 Show that, to 2nd order, allocative efficiency generates a redistribution-stabilization tradeoff when Pareto weights differ from income shares.

Environment

- The framework of this paper sits at the intersection of Baqaee and Farhi (2024) and La'O and Tahbaz-Salehi (2022).
- Static New Keynesian economy with C countries and N industries.
- Each country has a representative household that supplies a **distinct type of labor**.
- Countries can differ in labor supply elasticities, sectoral labor input shares, ownership structures, and consumption baskets.

Firms

- A unit mass of monopolistically competitive firms $k \in [0,1]$ in sector i

$$y_{ik} = z_i F_i(\{x_{ij,k}\}_{j=1}^{N}, \{L_{ic}\}_{c=1}^{C})$$

  - F_i is CRS, e.g. $F_i(\{x_{ij,k}\}_{j=1}^{N}, \{L_{ic}\}_{c=1}^{C}) = \varsigma_i \left(\sum_{j=1}^{N} \omega_{ij}^{\frac{1}{\sigma_i}} x_{ij,k}^{\frac{\sigma_i - 1}{\sigma_i}} + \sum_{c=1}^{C} \alpha_{ic}^{\frac{1}{\sigma_i}} L_{ic,k}^{\frac{\sigma_i - 1}{\sigma_i}} \right)^{\frac{\sigma_i}{\sigma_i - 1}}$.
  - In the calibration (geographic segmentation), α_{ic} equals the value-added share of industry i if it is located in country c and zero otherwise.
- Differentiated varieties within industry i are aggregated according to

$$y_i = \left(\int_0^1 y_{ik}^{\frac{\theta_i - 1}{\theta_i}} dk \right)^{\frac{\theta_i}{\theta_i - 1}}$$

- All firms in industry i minimize

$$mc_i = \min_{\{x_{ij,k}\}, \{L_{ic,k}\}} \sum_{c=1}^{C} w_c L_{ic,k} + \sum_{j=1}^{N} p_j x_{ij,k}, \quad s.t.\ y_{ik} = 1$$

Nominal Rigidities

- Firm k in industry i can reset price with probability $\delta_i \in (0, 1]$. The optimal reset price maximizes the profit

$$\Pi_{ik} = \max_{p_{ik}} [p_{ik} - (1 - \tau_i) mc_i] \left(\frac{p_{ik}}{p_i} \right)^{-\theta_i} y_i$$

- To a first-order approximation,

$$\log \boldsymbol{p} = \boldsymbol{\delta} \log \boldsymbol{mc}, \qquad \log \boldsymbol{\mu} = -(\boldsymbol{\delta}^{-1} - I) \log \boldsymbol{p}$$

where $\boldsymbol{\delta} = \text{diag}(\delta_1, ..., \delta_N)$.

Households

- Household c has KPR preferences

$$U_c(C_c, L_c) = \log(C_c) - V_c(L_c)$$

where C_c is a CRS aggregator of sectoral goods,

$$C_c = \mathcal{D}_c(\{c_{ci}\}_{i=1}^N).$$

For example, with CES preferences, $\mathcal{D}_c(\{c_{ci}\}_{i=1}^N) = \zeta_c \left(\sum_{i=1}^N \beta_{ci}^{\frac{1}{\sigma_c}} c_{ci}^{\frac{\sigma_c - 1}{\sigma_c}} \right)^{\frac{\sigma_c}{\sigma_c - 1}}$.

- Budget constraint

$$\text{GNE}_c \equiv \sum_{i=1}^N p_i c_{ci} \leq w_c L_c + \sum_{i=1}^N \Phi_{ic} \Pi_i - T_c$$

where Φ is the ownership matrix (geographic segmentation: $\Phi_{ic} = \mathbf{1}\{i \in N_c\}$).

Policy Instruments

- **Fiscal policy**: industry subsidies $1 - \tau_i = (\theta_i - 1)/\theta_i$ eliminate CES distortions; financed by lump-sum taxes proportional to ownership shares,

$$T_c = \sum_{i \in N} \Phi_{ic} \tau_i m c_i \int_0^1 y_{ik} dk.$$

- **Monetary policy**: controls nominal aggregate demand

$$\text{GNE} = \sum_{c=1}^{C} \sum_{i=1}^{N} p_i c_{ci} = m.$$

Equilibrium Definition

Definition

A *sticky-price equilibrium* is a set of allocations, prices, and policies such that:

(i) firms optimally choose inputs and reset prices when allowed;

(ii) each household maximizes utility subject to its budget constraint;

(iii) the government budget constraint is satisfied;

(iv) all markets clear.

Definition

The *flexible-price equilibrium* is defined by the same conditions, except that all firms reset their prices optimally.

- Log deviation from the flexible-price equilibrium: $\hat{x} = \log x - \log x^*$.

Minimum Notations

	Country c	**Union**
Income share	$\chi_c = \frac{\text{GNE}_c}{\text{GNE}}$	$\sum_c \chi_c = 1$
Consumption share	$\beta_{ci} = \frac{p_i c_{ci}}{\text{GNE}_c}$	$b_i = \sum_c \chi_c \beta_{ci}$
Domar weight	$\lambda_i^c = \sum_j \beta_{cj} \Psi_{ji}$	$\lambda_i = \sum_c \chi_c \lambda_i^c$

- $\Psi = (I - \Omega)^{-1}$ is the Leontief inverse, with input–output matrix $\Omega = [\omega_{ij}]$, $\omega_{ij} = \frac{p_j x_{ij}}{m c_i y_i}$.
- β_{ci} is **direct** exposure of country c to industry i.
- λ_i^c is **direct** and **indirect** exposure of country c to industry i.

Allocative Efficiency

Proposition

To a first-order approximation,

$$\hat{c}_c - \hat{l}_c = \underbrace{\mathcal{J}_c^{\mathrm{DI}\prime} \log \boldsymbol{\mu}}_{\textbf{direct incidence}} + \underbrace{\mathcal{J}_c^{\mathrm{FToT}\prime} \log \boldsymbol{\mu}}_{\textbf{factoral terms of trade}} \equiv \mathcal{J}_c' \log \boldsymbol{\mu}.$$

- **Direct incidence** reflects a country's dual role as both a consumer and a profit owner (supplier):

$$\Delta \mathrm{DI}_c \equiv \sum_i \left(\frac{\lambda_i \Phi_{ic}}{\chi_c} - \lambda_i^c \right) \log \mu_i = \mathcal{J}_c^{\mathrm{DI}\prime} \log \boldsymbol{\mu}$$

- **Factor terms of trade** measures the relative value of the labor a country supplies to the labor costs it faces:

$$\Delta \mathrm{FToT}_c \equiv d \log w_c - \sum_f \Lambda_f^c d \log w_f = \mathcal{J}_c^{\mathrm{FToT}\prime} \log \boldsymbol{\mu}$$

- The expression for $\mathcal{J}_c^{\mathrm{DI}}$ is **independent** of the specific forms of V_c, F_i, and $\mathcal{D}_c$.

Allocative Efficiency as Overall Terms of Trade

- In a horizontal economy (without input–output linkages), and assuming ownership segmentation,

$$\Delta \mathrm{DI}_c = \mathcal{J}_c^{\mathrm{DI}\prime} \log \boldsymbol{\mu} = \underbrace{\sum_{i \in N_c} \sum_{f \neq c} \chi_f \beta_{fi} / \chi_c \log \mu_i}_{\textbf{change in export prices}} - \underbrace{\sum_{i \notin N_c} \beta_{ci} \log \mu_i}_{\textbf{change in import prices}} .$$

- **The direct incidence effect** maps to the commodity terms of trade effect.
- **Allocative efficiency** = overall (commodity + factoral) terms of trade effect.

The Role of Production Networks

Input–Output Multiplier

$$\xi_c \equiv \sum_{i \in N_c} \frac{\lambda_i}{\chi_c} = \frac{\sum_{i \in N_c} p_i y_i}{\mathrm{GNE}_c}$$

measures total domestic sales relative to domestic final demand.

- Under ownership segmentation,

$$\Delta \mathrm{DI}_c = \underbrace{(\xi_c - \mathbb{E}_\chi[\xi])\,\overline{\log \mu}}_{\text{IO-multiplier channel}} + \underbrace{\left(\mathbb{E}_\chi[\xi] - \sum_i \lambda_i^c\right)\overline{\log \mu}}_{\text{home-bias channel}} + \underbrace{\sum_i \mathcal{J}_{ic}^{\mathrm{DI}}(\log \mu_i - \overline{\log \mu})}_{\text{sectoral-heterogeneity channel}}.$$

- Without production networks,

$$\xi_c = 1, \qquad \sum_i \lambda_i^c = 1,$$

so the first two channels disappear.

- When the IO-multiplier channel dominates, countries with above-average ξ_c experience a deterioration in their terms of trade following a monetary expansion ($\overline{\log \mu} < 0$).

Aggregate Cancellation: A Macro Envelope Theorem

Corollary

Income-share-weighted allocative efficiencies exactly cancel at first order:

$$\sum_{c=1}^{C} \chi_c(\hat{c}_c - \hat{l}_c) = \sum_{c=1}^{C} \chi_c \mathcal{T}_c' \log \boldsymbol{\mu} = 0.$$

- **Intuition:** a closed economy has zero aggregate terms of trade.
- **Implication:** monetary policy has a redistribution motive whenever welfare aggregation departs from income-share weighting.

Specification for the Second-Order Approximation

- Labor disutility function with household-specific Frisch elasticity η_c

$$V_c(L_c) = \psi_c \frac{L_c^{1+1/\eta_c}}{1+1/\eta_c}$$

- Across-sector elasticities are Cobb-Douglas

$$y_{ik} = z_i \varsigma_i \prod_{j=1}^{N} x_{ij,k}^{\omega_{ij}} \prod_{c=1}^{C} L_{ic,k}^{\alpha_{ic}}$$

and

$$C_c = \prod_{i=1}^{N} (c_{ci}/\beta_{ci})^{\beta_{ci}}.$$

Bergson–Samuelson Welfare

- Adopt the Bergson–Samuelson welfare aggregator

$$W(\{\kappa_c\}_{c=1}^C) = \sum_{c=1}^C \kappa_c U_c,$$

where $\{\kappa_c\}$ are Pareto weights satisfying $\sum_c \kappa_c = 1$.

Proposition (Bergson–Samuelson Welfare function)

For arbitrary Pareto weights $\{\kappa_c\}_{c=1}^C$,

$$W(\{\kappa_c\}) - W^*(\{\kappa_c\}) = \sum_{c=1}^C \kappa_c \mathcal{J}_c' \log \boldsymbol{\mu} - \frac{1}{2} \log \boldsymbol{\mu}' \mathcal{L}(\{\kappa_c\}) \log \boldsymbol{\mu}.$$

Details

- Generalizes Baqaee and Farhi (2024) (income-share weighting) and La'O and Tahbaz-Salehi (2022) (representative agent) to a HAIO economy with arbitrary Pareto weights.
- Under income-share weighting, monetary policy has a purely efficiency objective, $\sum_c \chi_c \mathcal{J}_c' \log \boldsymbol{\mu} = 0$.
- **Takeaway**: monetary policy faces a tradeoff between **aggregate stabilization** and **first-order redistribution** whenever Pareto weights deviate from income shares.

Optimal Policy

Proposition (Optimal Monetary Policy (Second Best))

Given Pareto weights $\{\kappa_c\}_{c=1}^{C}$*, the optimal monetary policy targets the sectoral price index*

$$\sum_{j=1}^{N} \zeta_j^*(\{\kappa_c\}) \log p_j = \pi^*(\{\kappa_c\})$$

with sectoral weights and optimal inflation bias

$$\zeta_j^*(\{\kappa_c\}) = (\delta_j^{-1} - 1) \sum_{i=1}^{N} (\delta_i^{-1} - 1) \varrho_i^m \mathcal{L}(\{\kappa_c\})_{ij},$$

$$\pi^*(\{\kappa_c\}) = \sum_{c=1}^{C} \kappa_c \mathcal{J}_c'(I - \boldsymbol{\delta}^{-1}) \varrho^m.$$

Here, $\varrho^m \equiv \frac{d \log \boldsymbol{p}}{d \log m} = \left[I - \varrho^w (\Gamma + \boldsymbol{\ell}^\mu \boldsymbol{\eta}^{-1})' (I - \boldsymbol{\delta}^{-1}) \right]^{-1} \varrho^w \mathbf{1}$ *denotes sectoral monetary pass-through.*

Centralized vs. Unilateral Optimal Policy

	Centralized	**Unilateral**
Pareto weights	$\kappa = \chi$	$\kappa = \mathbf{e}_c$ $(\kappa_c = 1)$
Welfare function	$-\frac{1}{2}\log\boldsymbol{\mu}'\mathcal{L}(\{\chi_c\})\log\boldsymbol{\mu}$	$\mathcal{J}_c'\log\boldsymbol{\mu} - \frac{1}{2}\log\boldsymbol{\mu}'\mathcal{L}(\{\mathbf{e}_c\})\log\boldsymbol{\mu}$
Objective	Minimize union-wide welfare loss	Minimize country c's welfare loss
Optimal policy	$\sum_j \zeta_j^*(\chi)\log p_j = 0$	$\sum_j \zeta_j^*(\mathbf{e}_c)\log p_j = \pi^*(\mathbf{e}_c)$
Inflation target	Zero	$\pi^*(\mathbf{e}_c)/\sum_j \zeta_j^*(\mathbf{e}_c)$

- $\pi^*(\mathbf{e}_c)$ measures how country c's **terms of trade** respond to a monetary expansion.
- Country c's unilateral optimal policy is inflationary if $\pi^*(\mathbf{e}_c) > 0$, indicating that its terms of trade improve following a monetary expansion.

Unilateral Inflation Stance

- The unilateral optimal policy
$$(\zeta^*(\mathbf{e}_c), \pi^*(\mathbf{e}_c)) = \arg\min_{(\zeta,\pi)} \mathbb{L}_c(\zeta, \pi)$$
assumes country c fully controls monetary policy.
- In a currency union, however, the target price index ζ is determined collectively (e.g., by the ECB Governing Council). Each country can only express its preferred inflation rate.

Definition

Given the union's target price index ζ, country c's **inflation stance** is

$$\pi_c(\zeta) = \arg\min_{\pi\in\mathbb{R}} \mathbb{E}[\mathbb{L}_c(\zeta, \pi)].$$

Proposition

The unilateral inflation stance satisfies

$$\pi_c(\zeta) = \underbrace{\frac{\zeta' \varrho^m}{[\zeta^*(\mathbf{e}_c)]' \varrho^m}}_{\text{price-index adjustment}} \pi^*(\mathbf{e}_c).$$

Interpretation. $\pi_c(\zeta)$ is a rescaled version of $\pi^*(\mathbf{e}_c)$.

Decomposition of Inflation Stance

- Recall that the unilateral inflation bias is

$$\pi^*(\mathbf{e}_c) = \mathcal{J}_c'(I-\boldsymbol{\delta})\boldsymbol{\varrho}^m$$

- The unilateral inflation stance decomposes into

$$\begin{aligned}\pi_c(\zeta) &= \frac{\zeta'\boldsymbol{\varrho}^m}{[\zeta^*(\mathbf{e}_c)]'\boldsymbol{\varrho}^m}\mathcal{J}_c'(I-\boldsymbol{\delta})\boldsymbol{\varrho}^m \\ &= \underbrace{\frac{\zeta'\boldsymbol{\varrho}^m}{[\zeta^*(\mathbf{e}_c)]'\boldsymbol{\varrho}^m}(\mathcal{J}_c^{\mathrm{DI}})'(I-\boldsymbol{\delta})\boldsymbol{\varrho}^m}_{\Delta\ \mathrm{DI}_c} + \underbrace{\frac{\zeta'\boldsymbol{\varrho}^m}{[\zeta^*(\mathbf{e}_c)]'\boldsymbol{\varrho}^m}(\mathcal{J}_c^{\mathrm{FToT}})'(I-\boldsymbol{\delta})\boldsymbol{\varrho}^m}_{\Delta\ \mathrm{FToT}_c}\end{aligned}$$

- Furthermore, within direct incidence (DI), into IO-multiplier, home-bias, and sectoral-heterogeneity channels.

Policy-Alignment Loss (PAL)

Definition (Policy-Alignment Loss)

The *policy-alignment loss* for country c under a union-wide policy (ζ, π) is

$$\mathrm{PAL}_c(\zeta, \pi) \;\equiv\; \mathbb{L}_c(\zeta, \pi) - \mathbb{L}_c\Big(\zeta^*(\mathbf{e}_c), \pi^*(\mathbf{e}_c)\Big) \;\geq\; 0.$$

Proposition (PAL Decomposition)

The expected PAL admits the closed-form decomposition

$$\mathbb{E}[\mathrm{PAL}_c(\zeta, \pi)] = \underbrace{\tfrac{1}{2}\,\frac{\zeta^*(\mathbf{e}_c)'\,\varrho^m}{(\zeta'\,\varrho^m)^2}\,[\pi_c(\zeta) - \pi]^2}_{\text{inflation misalignment}} + \underbrace{\mathcal{I}_c(\zeta)}_{\text{price-index misalignment}},$$

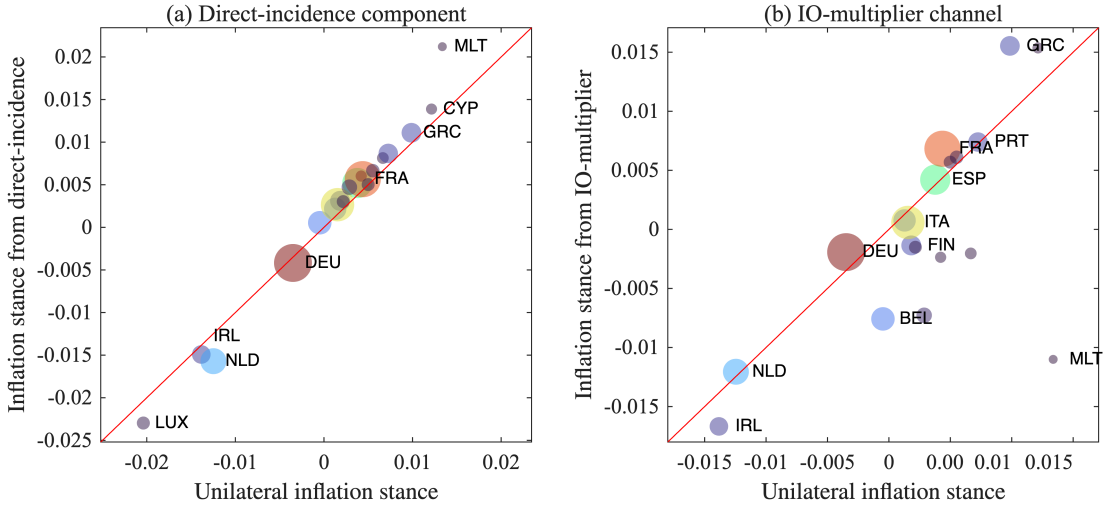
where $\mathcal{I}_c(\zeta) \geq 0$ *vanishes at* $\zeta = \zeta^*(\mathbf{e}_c)$.

- Inflation misalignment is **quadratic** in the distance $|\pi_c(\zeta) - \pi|$.

Calibration

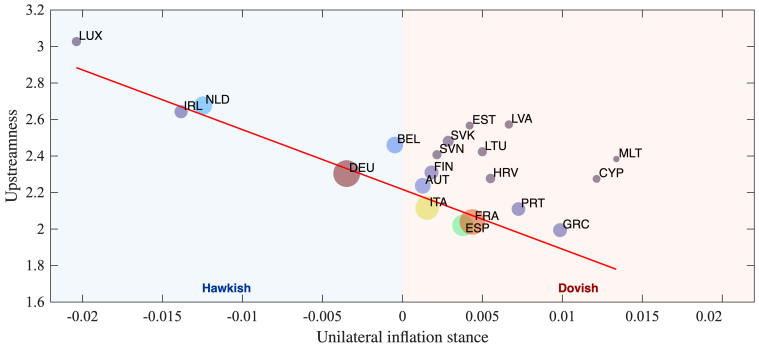
- **Parameters**: $\eta_c = 2$, $\theta_i = 6$, $\Phi_{ic} = \mathbf{1}\{i \in N_c\}$ (geographically segmented ownership).
- **IO matrix**: WIOD 2014 (Timmer et al., 2015); $C = 20$ euro-area countries, each with $N_c = 56$ industries.
- **Price rigidities**: monthly FPA from Pastén et al. (2020), mapped to WIOD industries via NAICS; converted to quarterly.
- **Sectoral productivity shocks**: annual TFP growth from WIOD SEA via Törnqvist index, interpolated to quarterly, linearly detrended, and used to form the variance–covariance matrix.

Decomposition of the Inflation Stance



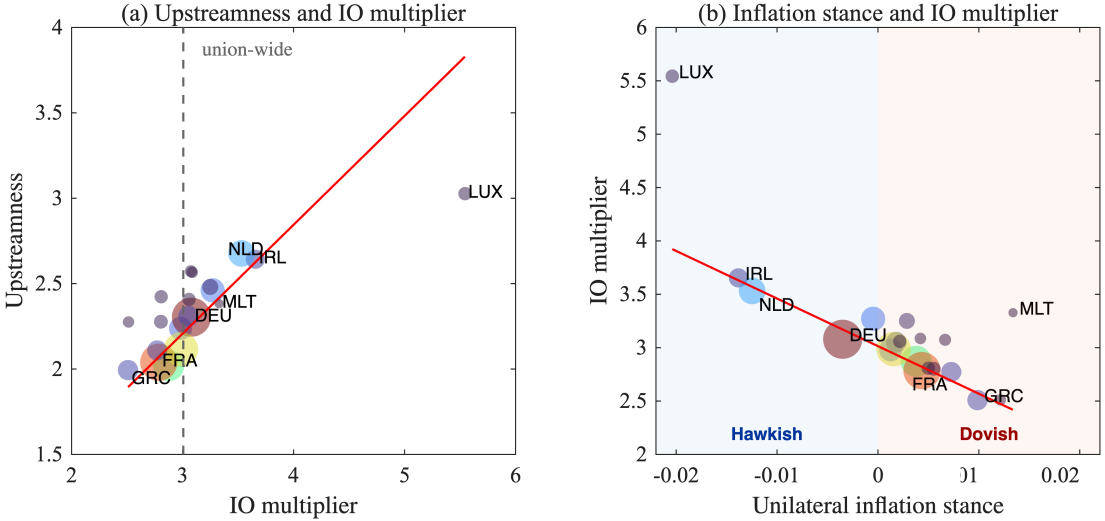
- Panel (a): The direct-incidence component reproduces almost the entire unilateral inflation stance.
- Panel (b): the *IO-multiplier channel alone* reproduces most cross-country variation.

Inflation Stance and Upstreamness



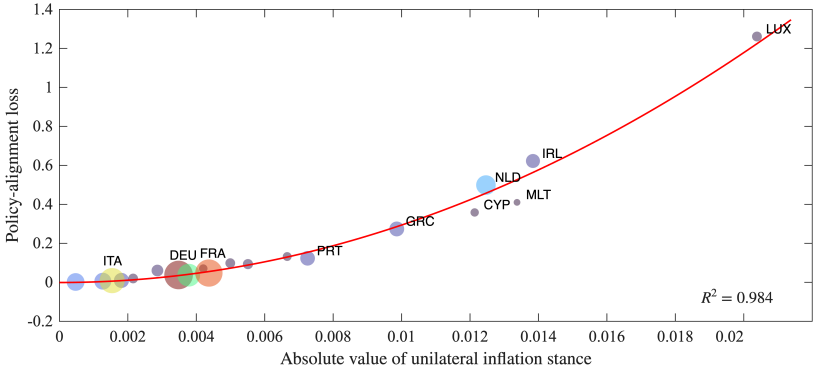
- **Upstreamness** (Antràs et al., 2012): value-added-weighted distance from final consumption.
- Strong negative relationship: WLS slope ≈ -32.7, $R^2 = 0.77$.
- More upstream countries prefer lower inflation; downstream prefer higher.
- Mirrors the familiar North–South divide in the euro area.

IO Multiplier as Mediating Network Statistic



- Panel (a): upstreamness ↔ local IO multiplier ($R^2 = 0.77$).
- Panel (b): IO multiplier ↔ unilateral inflation stance ($R^2 = 0.75$).
- Mechanism: network position ⇒ IO multiplier ⇒ terms of trade ⇒ inflation preference.

Inflation-Stance Deviation and PAL: Quadratic Fit



- Union-wide inflation consensus is zero under the centralized optimal policy.
- Quadratic fit (no linear term) gives $R^2 = 0.98$.
- A one-percentage-point inflation deviation predicts a loss of about 0.29 pp of steady-state consumption.
- Tail countries (LUX, IRL, NLD, GRC) bear disproportionate welfare costs.

Conclusion

- This paper develops a HAIO model and provides microfoundations for divergent inflation preferences across member states in a currency union.
- Bringing the model to the euro area, I find that a country's inflation stance negatively relates to it upstreamness, and the deviation from the union-wide consensus predicts welfare costs.
- Therefore, countries at either extreme of the production network, whether highly upstream or highly downstream, incur the largest welfare costs.

Work in Progress

- Country level misallocation through global production networks
- Optimal tariff policy with strategic complementarities in global production networks
- Aggregation and skewness in production networks
- Industry policy and entry cascades in production networks (with Kaiji Chen)
- ...

Appendix

Lemma

The second-order welfare loss under arbitrary weights $\{\kappa_c\}$ is $\mathcal{L}(\{\kappa_c\}) = \sum_{c=1}^{C} \kappa_c \mathcal{L}_c$ where $\mathcal{L}_c = \mathcal{L}_c^{\text{e.g.}} + \mathcal{L}_c^{\text{within}} + \mathcal{L}_c^{\text{across}}$ summarizes three sources of misallocation:

$$\begin{aligned}
\mathcal{L}_c^{\text{e.g.}} &= \tfrac{\eta_c}{1+\eta_c} \chi_c^{-2} [\lambda \circ \Phi_{(:,c)}][\lambda \circ \Phi_{(:,c)}]', \\
\mathcal{L}_c^{\text{within}} &= \text{diag}(\lambda^c \circ \theta)(\delta^{-1} - I)^{-1}, \\
\mathcal{L}_c^{\text{across}} &= \tfrac{\eta_c}{1+\eta_c} \Upsilon^c + \sum_{f=1}^{C} \tfrac{1}{1+\eta_f} \Lambda_f^c \Upsilon^f - \Xi^c + \sum_{f=1}^{C} \Lambda_f^c \Xi^f,
\end{aligned}$$

with income-share and labor-wedge Hessians Ξ_{ij}^c and Υ_{ij}^c

$$\Xi_{ij}^c \equiv \tfrac{\partial^2 \log \chi_c}{\partial \log \mu_i \partial \log \mu_j} = -\Gamma_{jc}\Gamma_{ic} + \chi_c^{-1} \sum_{f=1}^{C} Q_{cf}(\Phi_{if} - \Psi_{if}) \Big(2 \sum_{g=1}^{C} \lambda_i^g \chi_g \Gamma_{jg} + \lambda_i \iota_{ij} - 2\lambda_j \Psi_{ji}\Big),$$

$$\Upsilon_{ij}^c \equiv \tfrac{\partial^2 \log(\Lambda_c/\chi_c)}{\partial \log \mu_i \partial \log \mu_j} = \tfrac{\Phi_{ic}}{\chi_c} \Big(2\lambda_j \Psi_{ji} - 2 \sum_{f=1}^{C} \lambda_i^f \chi_f \Gamma_{jf} + 2\lambda_i \Gamma_{jc} - \lambda_i \iota_{ij} - \lambda_i \lambda_j \tfrac{\Phi_{jc}}{\chi_c}\Big).$$

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